Théo BEILLE

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Summary

MSc MVA (ENS Paris-Saclay) and ENSTA Paris (applied mathematics, top 5%) student, with research experience in learning-to-optimise. **Seeking a 6-month research internship from April 2027** in machine learning / computer vision.

Research Experience

Visiting Researcher – KTH Royal Institute of Technology *Jan. – Jun. 2026*
Supervisor: Prof. Ozan Öktem – Paper in preparation *Stockholm, Sweden*

- Designed a learned optimisation method: a convolutional neural network (deep unrolling) that predicts the acceleration parameters of an iterative optimisation algorithm, combined with a mathematical safeguard that preserves the algorithm's convergence guarantees.
- **Result:** reconstruction up to **10x more accurate** than the classical algorithm after about 10 iterations, evaluated on real clinical CT data (512x512 images).
- Owned the full research loop: literature review, method design, PyTorch implementation, experimental protocol, benchmarking against classical baselines, evaluation metrics and written report.
- Released the code openly: github.com/TheoBeille/learned_deviations_tgv2. **Stack:** Python, PyTorch, NumPy, Git.

Education

MSc MVA (Mathematics, Vision, Learning) – ENS Paris-Saclay *2026 – 2027*
Courses: Deep Learning, Computer Vision, Reinforcement Learning, Graphs in Machine Learning, ALTEGRAD (Advanced Learning for Text and Graph Data), Computational Statistics.

ENSTA Paris – Institut Polytechnique de Paris *2024 – 2027*
Engineering degree (MSc), Applied Mathematics – Top 5% (216 students), GPA 4.17/4.30 *Palaiseau, France*
Courses: optimisation, probability, statistics, Bayesian filtering, nonlinear time series, functional analysis.

KTH Royal Institute of Technology – Erasmus Exchange, Applied Mathematics *Jan. – Jun. 2026*
Courses: nonlinear optimisation, machine learning for stochastic differential equations, stochastic calculus, time series analysis.

Lycée Pierre Corneille – Preparatory Classes (CPGE MPSI / MP*) *2022 – 2024*
Intensive two-year programme in mathematics and physics, preparing for the national competitive exams to French engineering schools.

Leadership and Teaching

Head of Logistics – Trium ENSTA Paris (career fair) *2025 – 2026*

- Organised one of the largest student career fairs in France (160+ companies, 3,000 students, budget over €200,000), leading a team of volunteers from set-up to the event itself.

Mathematics Oral Examiner (kholleur) – CPGE PSI *2026 – present*
Weekly oral mathematics examinations for preparatory-class students.

Skills

Programming: Python, C++
Machine learning: deep learning, convolutional neural networks, deep unrolling, learned optimisation, statistical learning, probabilistic modelling, time series
Mathematics: optimisation (nonlinear, iterative methods), inverse problems, probability, statistics, stochastic calculus, functional analysis
Libraries and tools: PyTorch, NumPy, Pandas, SciPy, Statsmodels, Git, GitHub, LaTeX
Languages: French (native), English (C1) | **Interests:** sailing, kitesurfing, swimming, piano